

**LITHIUM-THIONYL CHLORIDE BATTERIES
FOR THE
MARS PATHFINDER MICROROVER**

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AGENDA

- **BACKGROUND - MISSION DESCRIPTION**
- **BATTERY REQUIREMENTS**
- **CELL DESIGN**
- **BATTERY DESIGN**
- **TEST DESCRIPTIONS & RESULTS**
- **SUMMARY**

BACKGROUND

- MARS PATHFINDER MISSION CONSISTS OF THE LANDER AND THE MICROROVER - ONE OF THE FIRST *FASTER-BETTER-CHEAPER* NASA/JPL MISSIONS
- MICROROVER IS CONSIDERED THE EXPERIMENT ON THE MARS PATHFINDER MISSION
- MAIN POWER PROVIDED TO MICROROVER BY SOLAR ARRAY
- LITHIUM PRIMARY BATTERIES SUPPORT
 - KEEP ALIVE CIRCUITS DURING THE NIGHT
 - DURING DUST STORMS
 - LOAD LEVELING WHEN WHEEL IS STUCK
- BATTERIES ARE ENCLOSED IN THE *Warm Electronic Box (WEB)*
 - BATTERY TEMPERATURE IS PROJECTED BETWEEN -30° C & +30° C

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BATTERY REQUIREMENTS

- CLASS 'D' MISSION
- LITHIUM THIONYL CHLORIDE CHEMISTRY
- SIZE/CONFIGURATION = 'D' SIZE / SPIRAL WOUND
- THREE CELLS IN SERIES / THREE BATTERIES IN PARALLEL
- WEIGHT ALLOCATION:
 - CELLS = 1080g
 - WIRE & DIODES = 45g
 - CASES = 54g
- MAX DISCHARGE RATE = 750 mA
- OPERATING TEMPERATURE = -30° to +30° C

BATTERY REQUIREMENTS con't

- MINIMUM CAPACITY @ -30° C = 6 Ah
- MINIMUM OPERATING CELL VOLTAGE @ -30° C & 750 mA = 2.5 V
- SAFE
- STORAGE MODE TEMPERATURES
 - GROUND STORAGE @ 5° C for 21 MONTHS
 - INTEGRATION @ 40° C for 3 MONTHS
 - CRUISE @ 30° C for 8 MONTHS
- SURVIVE LAUNCH-CRUISE-LANDING ENVIRONMENTS

CELL DESIGN

- LITHIUM ANODE
 - WEIGHT = 4.11 g (INCLUDING 0.48g FOLDBACK Li)
 - CAPACITY = 14 Ah
 - EXCESS CAPACITY (FOLDBACK) = 1.85 Ah
- CARBON CATHODE
 - CARBON MIX = 61% SHAWINIGAN ACETYLENE BLACK
20% EC 300J ACETYLENE BLACK
19% PTFE
- ELECTROLYTE
 - CAPACITY = 20 Ah
 - COMPOSITION = 1.35 M LiCl_4Al + 0.55 M SO_2 IN SOCl_2 & 3 ADDITIVES
- SEPARATOR = HESGON WOVEN GLASS ('M' WEAVE HEAT CLEANED)
- VENT = FUSITE 325 $\pm$ 50 psi

BATTERY DESIGN

- THREE 'D' SIZE CELLS IN SERIES (BATTERY)
- THREE BATTERIES IN PARALLEL (TOTAL NINE CELLS)
- ALUMINUM BATTERY CASE w/VENT HOLES
- CELLS WRAPPED IN KAPTON TAPE
- NON-CONDUCTIVE SPACERS BETWEEN CELLS
- ADHESIVE USED TO BOND CELLS TO ALUMINUM CASE
- EACH BATTERY HAS ONE DIODE AND ONE FUSE
- TWO HEATERS & ONE THERMISTOR PER BATTERY

BATTERY WEIGHT DISTRIBUTION

CELLS	9 x 118 g = 1062.0 g	85.4 %
BATTERY CASES	3 x 32.35g = 97.1 g	7.8 %
CELL SPACERS	4 x 4.35 g = 17.4 g	1.4 %
<u>OTHER</u>		
BONDING MATERIAL		
DIODES, FUSES		
HEATERS, PRTs		
CABLE, CONNECTOR		
KAPTON TAPE	66.3 g	5.4 %
	<u>TOTAL 1242.8 g</u>	

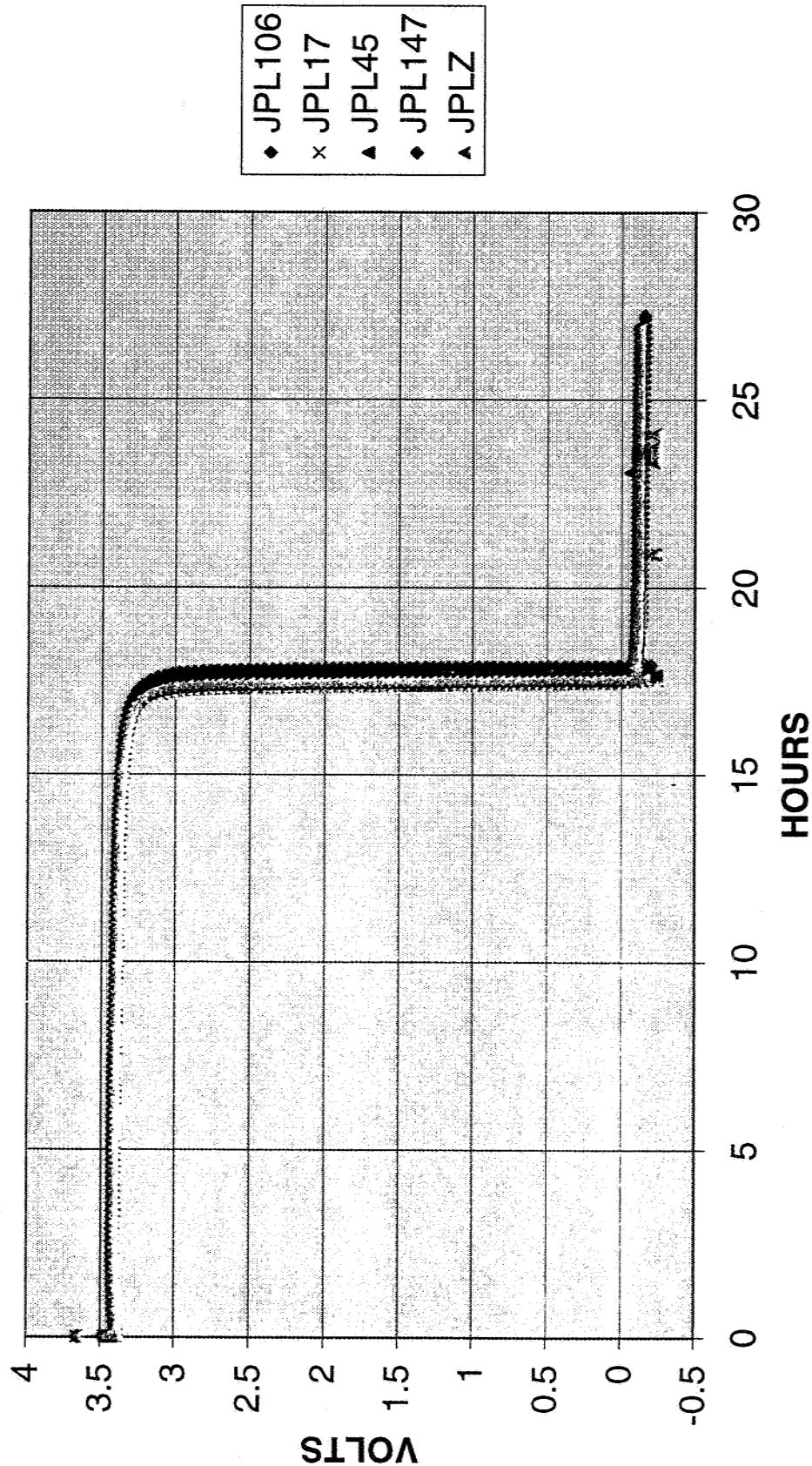
CELL/BATTERY LOT DESCRIPTION

- **SAFT WILL SUPPLY TWENTY-ONE 3-CELL BATTERIES & 25 CELLS**
 - **10 CELLS TESTED @ SAFT (5 @ R.Temp. & 5 @ -30° C)**
 - **15 CELLS TESTED @ JPL FOR ELECTRONIC COMPONENT CHECK**
 - **SIX 3-CELL 'TEST' BATTERIES (TESTED @ SAFT)**
 - **SIX 3-CELL 'SIM' (System Integration Model) BATTERIES TESTED ON 'SIM' MICROROVER @ JPL**
 - **THREE 3-CELL 'TEST & INTEGRATION' BATTERIES FOR TESTING & INTEGRATING FLIGHT UNIT @ JPL**
 - **THREE 3-CELL 'FLIGHT' BATTERIES**
 - **THREE 3-CELL 'FLIGHT SPARE' BATTERIES**

'TEST' BATTERY TEST PLAN

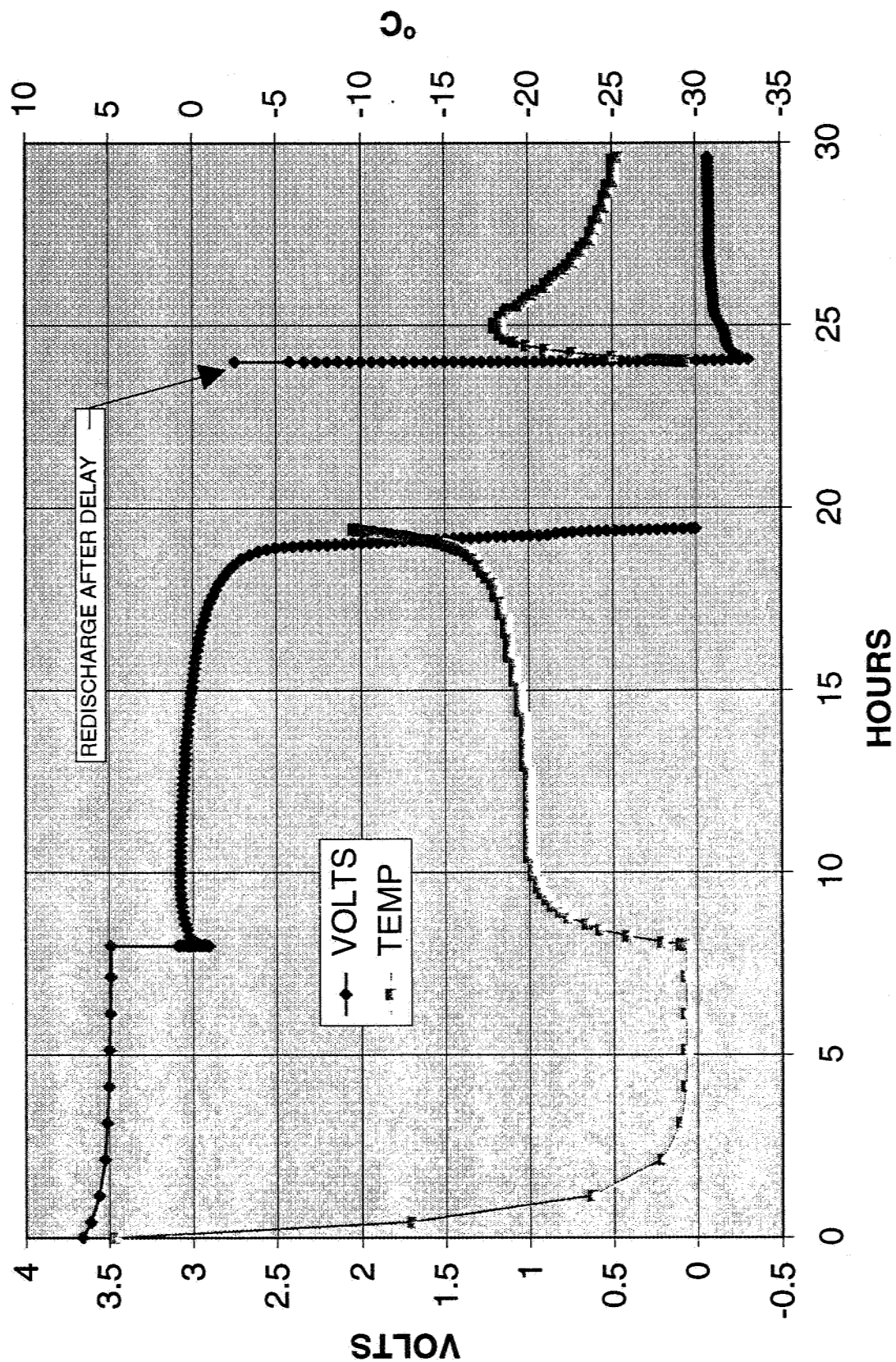
- **SIX 3-CELL 'TEST' BATTERIES (TESTED @ SAFT)**
 - **QUASI-STATIC ACCELERATION @ 66g for 1 MINUTE IN BOTH DIRECTIONS (AXIAL & RADIAL) ON A CENTRIFUGE (6 BATT.)**
 - **RANDOM VIBRATION IN BOTH DIRECTIONS (6 BATT.)**
 - **PYRO-SHOCK PULSE IN BOTH DIRECTIONS (6 BATT.)**
 - **35 THERMAL CYCLES BETWEEN -55° C & +60° C WITH 1 HOUR DWELL AT BOTH TEMPERATURES & 1 HOUR TRANSITION (6 BATT.)**
 - **2 BATTERIES DISCHARGED @ 750 mA & -30° C**
 - **1 BATTERY DISCHARGED @ 750 mA & 25° C**
 - **2 BATTERIES SIMULATE MARTIAN SURFACE OPERATIONS**
 - **1 BATTERY SHORT WITH FUSE REMOVED**

JPL 36/58 CELLS .75 A LOT ACCEPTANCE TEST **DISCHARGE AT ROOM TEMPERATURE** **5 CELLS, 1 WEEK OLD**



JPL147.XLS

JPL 36/58 CELL #52, .75 A LOT ACCEPTANCE TEST DISCHARGE AT -30 °C

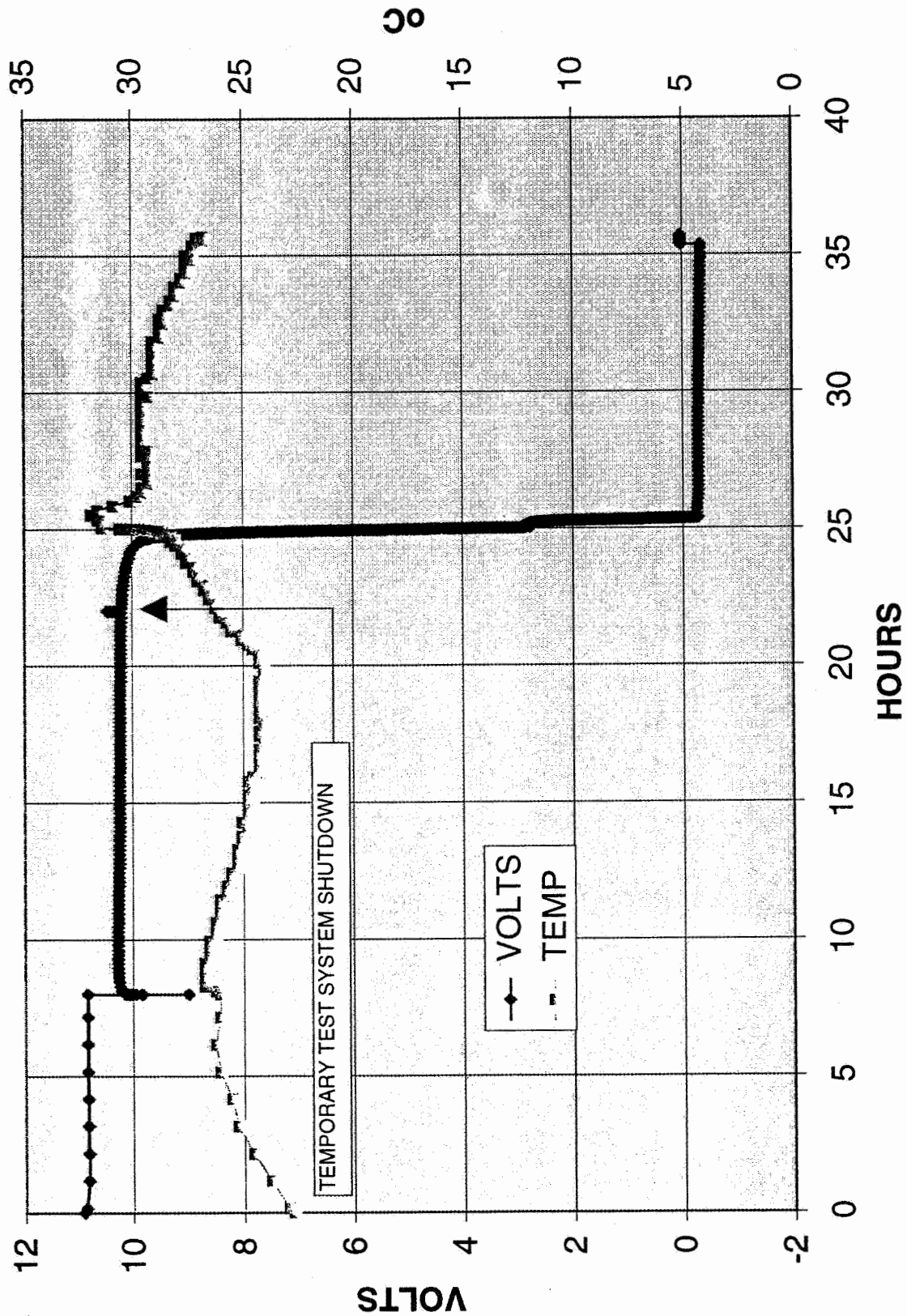


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Chart1

JPL52.XLS

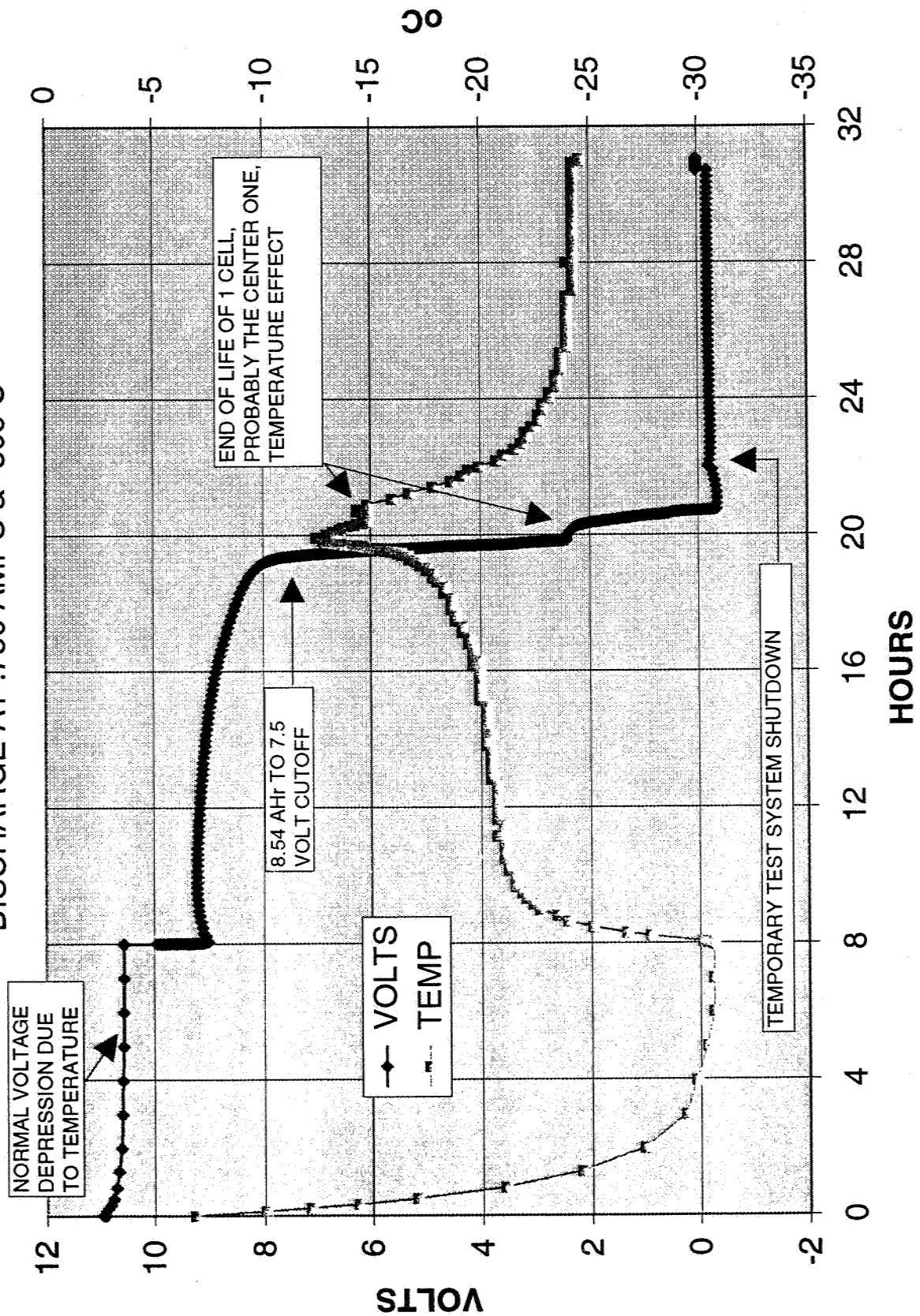
JPL BATTERY #6 **DISCHARGE AT .750 AMPS & ROOM TEMP.**



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JPLBATT.XLS

JPL BATTERY #5 **DISCHARGE AT .750 AMPS & -30o C**



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MARTIAN SURFACE BATTERY OPERATION

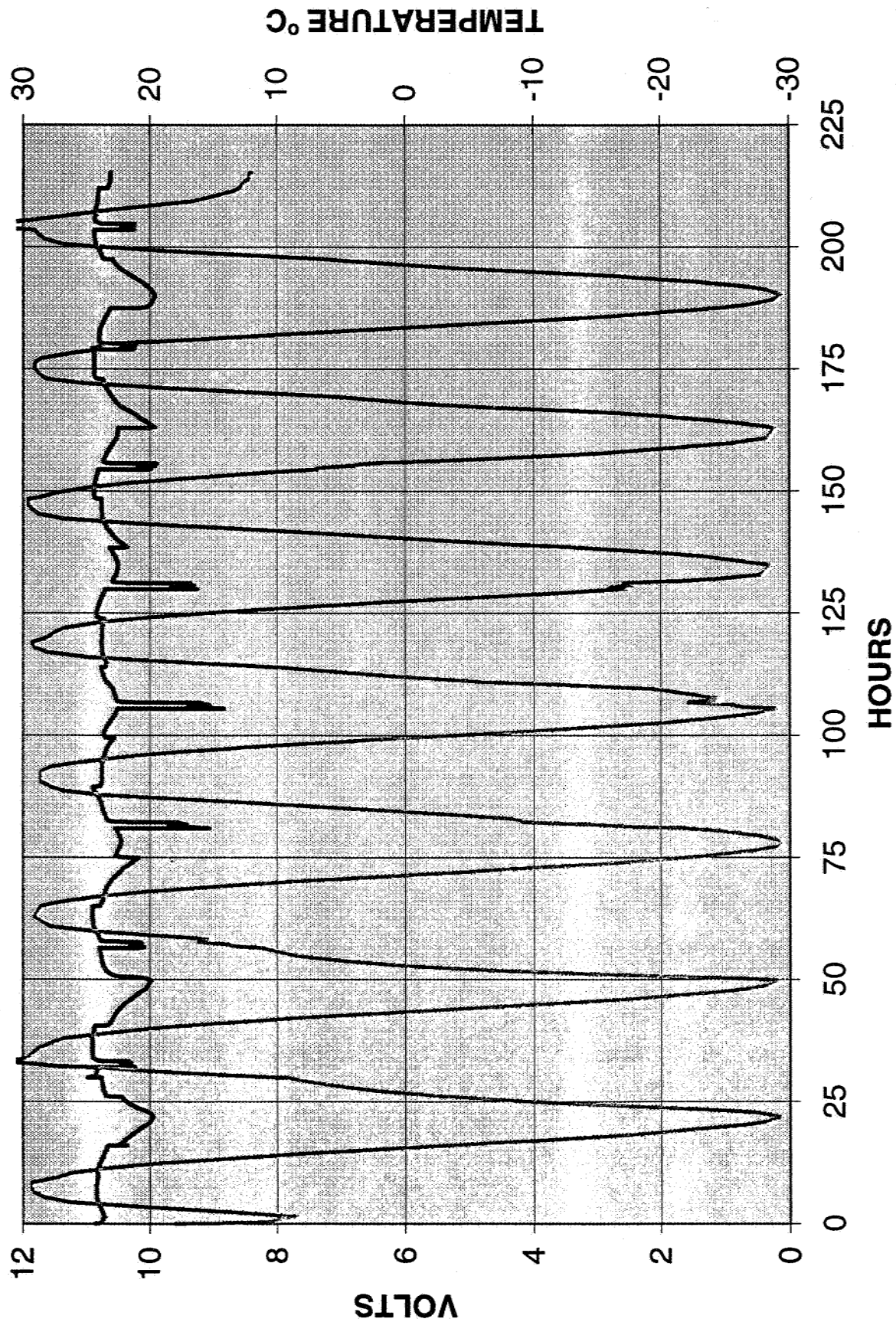
- 50 mA DISCHARGE FOR 10 HOURS (0.5 Ah)
- 2 mA DISCHARGE FOR 6 HOURS (0.012 Ah)
- 750 mA DISCHARGE FOR 15 MINUTES (0.1875 Ah)
- 400 mA DISCHARGE FOR 30 MINUTES (0.2 Ah)
- 750 mA DISCHARGE FOR 30 MINUTES (0.375 Ah)
- 2 mA DISCHARGE FOR 45 MINUTES (0.0015 Ah)
- OPEN CIRCUIT FOR 6.5 HOURS (0.0 Ah)

TOTAL / DAY (1.276 Ah)

REPEAT UNTIL BATTERY IS DEPLETED

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MARTIAN TEMPERATURE CYCLE BATTERY DISCHARGE(BATT #7)



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SUMMARY

- **Li-SOCl₂ PRIMARY BATTERY SELECTED FOR THE MARS PATHFINDER MICROROVER**
- **TEST BATTERIES & SIM BATTERIES DELIVERED & TESTED**
- **SPECIFIC ENERGY PENALTY**
 - FROM CELL TO BATTERY LEVEL WAS 15%
 - FROM ROOM TEMPERATURE TO -30° C WAS 45%
- **BATTERIES DELIVERED FROM SAFT AMERICA TO-DATE HAVE MET MARS PATHFINDER MICROROVER REQUIREMENTS**
- **CAN PROCEED WITH THE MANUFACTURING OF FLIGHT & FLIGHT SPARE UNITS**

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